

their “speech.” Certain key protocols you should know about are:

TCP: Transmission Control Protocol, used for the reliable transmission of data over a network.

IP: The Internet protocol is the set of rules that dictates how exactly packets of data should be transmitted over the Internet. IP standardises the way machines over the Internet (or any IP network) forward or route their packets based on their IP addresses.

HTTP: Hypertext Transfer Protocol, used for transmitting and displaying information in the form of Web pages on browsers. This is the language used when your Web browser talks to the Web server.

FTP: File Transfer Protocol, used for file transfer (uploading and downloading) over the Internet. Both FTP and HTTP are based on TCP/IP. FTP is still often used as a means of downloading large files.

SMTP: Simple Mail Transfer Protocol, used for e-mail.

Ethernet: Used for data transmission over a LAN (Local Area Network).

Wi-Fi: The wireless version of Ethernet.

Of all these protocols, in order to understand how the Internet works, what we really need to know is the IP and the TCP.

2.1.8 TCP/IP

Every device, every computer, that is part of the Internet, is allocated an address, called an Internet address, or Internet Protocol address (IP address). However, it isn’t like our home address or anything because, as we mentioned earlier, computers only understand numbers. So the IP address is a series of numbers.

The current protocol is called IPv4. Each IP address is actually just four numbers each ranging from 0 to 255 (each of which is called an “octet”) and separated by decimal points (called dots). IP addresses are, therefore in the format xxx.xxx.xxx.xxx where each xxx could be any number from 0 to 255. So an IP address could look something like 161.184.138.36. This address is essential during every exchange of data on the Internet, because it identifies the client computer (which makes the request, and to which the data must be sent) and the destination computer (to which the request must reach). Nothing, therefore, can happen on the Internet unless your computer is first